Exercise Sheet

Lecture 11

Dual Spaces, Functionals, and Multilinear Forms

A linear functional acts as $\varphi(v) = \varphi_i v^i$; the dual basis satisfies $e^i(e_j) = \delta^i_j$. A ⋆ marks a harder problem.

Dual basis and functionals

Exercise 11.1 (computational) Find the dual basis of $e_1 = (2,1)$, $e_2 = (1,1)$ in $\mathbb{F}^2$.

Exercise 11.2 (computational) On $\mathcal{P}_2$, express the functional $\varphi(a + bx + cx^2) = a + c$ in the dual basis of $1, x, x^2$.

Exercise 11.3 (computational) Find the dual basis of $e_1 = (1,0)$, $e_2 = (1,1)$ in $\mathbb{F}^2$.

Exercise 11.4 (computational) Find the dual basis of $e_1 = (3,1)$, $e_2 = (5,2)$ in $\mathbb{F}^2$.

Exercise 11.5 (computational) On $\mathcal{P}_2$ with basis $1, x, x^2$, expand $\varphi(a + bx + cx^2) = a - b + 2c$ in the dual basis and evaluate φ on $1 + 2x + 3x^2$.

Exercise 11.6 (proof) Prove that for a subspace $U \subseteq V$, $\dim U + \dim U^0 = \dim V$.

Exercise 11.7 (proof) Prove that the dual basis $e^1, \ldots, e^n$ is linearly independent in V^*.

The dual map

Exercise 11.8 (computational) Let $T\colon \mathbb{F}^2 \to \mathbb{F}^2$ have matrix $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$. Compute $\mathcal{M}(T')$ in the dual bases.

Exercise 11.9 (computational) Let $T\colon \mathbb{F}^2 \to \mathbb{F}^2$ have matrix $\left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$. Compute $\mathcal{M}(T')$ in the dual bases.

Exercise 11.10 (computational) Let $T\colon \mathbb{F}^3 \to \mathbb{F}^2$ have matrix $\left(\begin{smallmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{smallmatrix}\right)$. Write $\mathcal{M}(T')$ and state the domain and codomain of T'.

Exercise 11.11 (computational) Let $T\colon \mathbb{F}^2 \to \mathbb{F}^2$ have matrix $\left(\begin{smallmatrix} 2 & 0 \\ 1 & 3 \end{smallmatrix}\right)$. For $\varphi = 2e^1 - e^2$ compute $T'\varphi$ in the dual basis.

Exercise 11.12 (proof) Prove that $(ST)' = T'S'$ for $T\colon U \to V$, $S\colon V \to W$.

Exercise 11.13 (proof) Prove $\operatorname{null} T' = (\operatorname{range} T)^0$.

Exercise 11.14 (proof) Prove that $(S+T)' = S' + T'$ for $S, T\colon V \to W$.

Bilinear and multilinear forms

Exercise 11.15 (computational) Write the matrix of $B(u,v) = 2u_1v_1 - u_1v_2 + 3u_2v_2$, and split it into symmetric and antisymmetric parts.

Exercise 11.16 (computational) For B with matrix $\left(\begin{smallmatrix} 1 & 2 \\ 3 & 1 \end{smallmatrix}\right)$, $u = (1,1)$ and $v = (2,-1)$, compute $B(u,v) = u^{\mathsf{T}}Bv$.

Exercise 11.17 (computational) Write the matrix of $B(u,v) = u_1v_1 + 4u_1v_2 + 2u_2v_1 - u_2v_2$ and split it into symmetric and antisymmetric parts.

Exercise 11.18 (computational) Write the matrix of the symmetric bilinear form on $\mathbb{R}^2$ whose quadratic form is $Q(v) = v_1^2 + 4v_2^2 + 2v_1v_2$.

Exercise 11.19 (proof) Let $f\colon (\mathbb{F}^n)^n \to \mathbb{F}$ be multilinear in its n column arguments and suppose f vanishes whenever two columns are equal. Prove that f is then *alternating*: swapping two columns changes the sign. (The determinant is such an f, so this shows column swaps flip its sign.)

Exercise 11.20 (proof) Prove that every bilinear form B on V splits uniquely as $B = S + A$ with S symmetric and A alternating.

Covariant and contravariant

Exercise 11.21 (computational) In a basis with metric $g_{ij} = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$, lower the index of the vector with contravariant components $v^i = (1,2)$, and compute $\langle v, v\rangle$.

Exercise 11.22 (computational) With metric $g_{ij} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 4 \end{smallmatrix}\right)$, lower the index of $v^i = (2,1)$ and compute $\langle v, v\rangle$.

Exercise 11.23 (computational) With metric $g_{ij} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, lower the index of $v^i = (1,2)$ and compute $\langle v, v\rangle$.

Exercise 11.24 (computational) The metric $g_{ij} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has inverse $g^{ij} = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$. Raise the index of the covector $\varphi_i = (1,3)$.

Exercise 11.25 (computational) With the Minkowski metric $g_{\mu\nu} = \operatorname{diag}(1,-1,-1,-1)$ and $x^\mu = (2,1,0,3)$, find x_μ and the interval $x_\mu x^\mu$.

Exercise 11.26 ($\star$) Show that the canonical map $V \to V^{**}$, $v \mapsto (\varphi \mapsto \varphi(v))$, is injective (hence an isomorphism in finite dimensions).